

SANGJUN LEE

silee9920@gmail.com | <https://www.linkedin.com/in/sangiunnlee/> | <https://sangiunnlee.github.io/>

EDUCATION

University of California, Berkeley

May 2026

Bachelor of Science in Electrical Engineering and Computer Science

Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Designing Information Devices and Systems, Discrete Mathematics, Probability, Multivariable Calculus, Structure and Interpretation of Computer Programs, Data Science

Teaching: Great Ideas in Computer Architecture (Machine Structures) (uGSI & AI, CS 61C), The Foundations of Data Science (uGSI, Data 8)

SKILLS

Languages: Python, Java, C, JavaScript, HTML/CSS, Swift, SQL, RISC-V Assembly

Frameworks & Tools: Spring Boot, Linux, React, React Native, Node.js, Express.js, FastAPI, Docker, Git, Firebase, Google Cloud Platform, Shell, PyTorch, scikit-learn, NumPy, Pandas, OpenMPI, SIMD

Languages: Native Proficiency in English, Native Proficiency in Spanish, Native Proficiency in Korean

EXPERIENCE

Samsung SDS

JUNE 2025 - AUG 2025

Software Engineer Intern

South Korea

- Developed a GUI agent to automate complex workflows of over 14 sequential actions, leveraging Multimodal Large Language Models (MLLMs) to reduce manual task completion time by approximately 80%
- Engineered the core system architecture, including a Python-based action execution engine that translates AI-generated instructions into precise, automated on-screen actions (e.g., clicks, text entry)
- Boosted MLLM action-prediction accuracy by 30% through systematic prompt engineering and testing of over 20 distinct variations

Korean Air Force Headquarters

OCT 2023 - MAY 2025

Backend Developer

South Korea

- Developed and optimized the backend infrastructure for an automated crack detection system for runways using Spring Boot, reducing runway inspection and labeling time from over 100 hours to 2-3 hours
- Increased detection accuracy by 35% by deploying a FastAPI server with YOLOv8 & OpenCV that processes 4,000+ images per hour
- Engineered and secured an Oracle database for 100,000+ runway slabs, managing data access and permissions for 5,000+ users

Machine Learning Engineer

South Korea

- Recognized with the Prime Minister Award in Korea for developing an AI solution that reduced the time soldiers spent on administrative tasks from 20 to 3 hours weekly, impacting over 50,000 soldiers
- Trained, fine-tuned, and deployed transcription models via a FastAPI server, cutting manual transcription time from weeks to minutes for 200+ users and increasing throughput by 500%
- Streamlined development in a high-security, closed-network environment, improving deployment time by 200% and reducing application startup by 5 minutes using Docker for containerization

Great Ideas in Computer Architecture (Machine Structures)

JUNE 2023 - AUG 2023

Undergraduate Student Instructor

Berkeley, CA

- Instructed for a computer architecture class of 200+ students, leading to a 90% increase in student participation in discussion sections
- Created weekly discussion worksheets on core concepts such as C, RISC-V, hardware, and optimization (SIMD, OpenMP)

AlectoAI

JUNE 2023 - AUG 2023

Software Engineering Intern

Berkeley, CA

- Engineered scalable backend APIs in Swift for an AI-powered facial recognition platform, implementing core features like commenting and liking to enhance user engagement
- Managed and constructed the Firebase database schema to ensure efficient data storage and retrieval while maintaining performance

PROJECTS

Rook

OCT 2024 - MAY 2025

- Developed a full-stack web application using React and Express that recommends similar books based on a user's reading list
- Applied prompt engineering with OpenAI to generate book recommendations and used Google Books API for book metadata
- Built an authentication system with login/logout functionality and authorization with JWT, ensuring protected user connection

Gitlet

May 2022

- Developed a version-control system that mimics Git's core functionalities such as staging, committing, branching, and merging
- Incorporated serialization and hashing (SHA-1) to efficiently save files and commit trees, ensuring persistence across versions
- Implemented tree traversal algorithms to optimize performance while navigating through commit histories and branches